Critical Infrastructure Security

IY5612 (IY3612/IY4612) — Academic Year 2025/2026 (Unit 3)

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Unit 3— Robustness of Networks and Cascading Failures

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Models of Large-Scale Networks

Robustness of Networks

Robustness of Networks and Cascading Failures
- Characterising E-R Graphs
- Networks and Models
- The Watts-Strogatz Model
- Scale-Free Graphs
- The Barabási-Albert Model

Part II

Robustness of Networks

This block covers

- Properties and Metrics for Characterising Graphs
- Existing Networks and the Erdős-Rényi Model
- The Watts-Strogatz Small-World Model
- Scale-Free Graphs and the Barabási-Albert Model

Required Reading

- M. van Steen:
  Graph Theory and Complex Networks: An Introduction
  Maarten van Steen (self-published, 2010)
- M. E. J. Newman:
  Networks: An Introduction
  Oxford University Press, 2010

  Chapters 7 (Measures and Metrics), 8 (The Large-Scale Structure of Networks)

Characterising E-R Graphs

We continue with our exposition of graph characteristics beginning with the simplest case of Erdős-Rényi random graphs and moving along to more complex but realistic network models widely used to capture the properties and behaviour of large-scale networks such as the Internet

Erdős-Rényi Graph Connectivity (1/3)

When increasing p, most vertices are contained in the same component (this is called the giant component — a highly prominent feature of random graphs, where we can show a sharp parameter bound):

- If $p \leq \frac{1-\varepsilon}{n}$ for any $\varepsilon > 0$, the size of connected components will be similar on the order of $\log n$ with high probability
- For $p > \frac{1+\varepsilon}{n}$, a single giant component arises with high probability with other components of size $O(\log n)$
- For $p = \frac{1}{n}$ the number of vertices in the largest component of the graph is again with high probability proportional to $n^{\frac{2}{3}}$

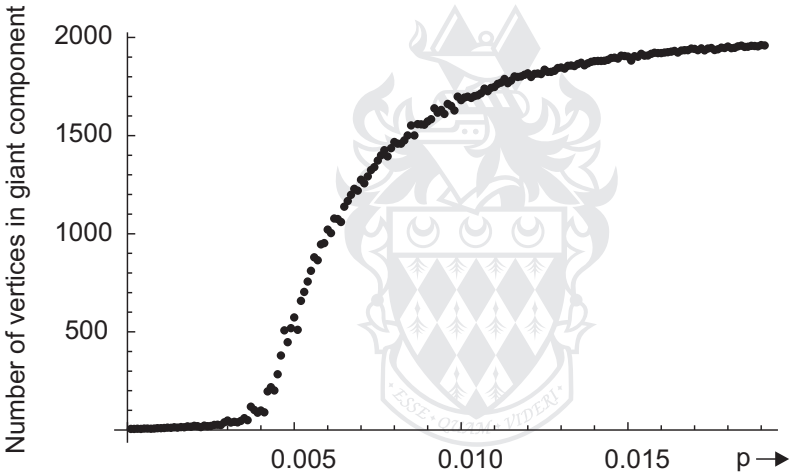
Average Path Lengths

Average Path Lengths in Erdős-Rényi Graphs

By looking at the average degree k, it is easier to observe an interesting property of Erdős-Rényi random graphs:

- $k < 1$ Clusters will be small and isolated, components will have small diameter, path lengths are short
- $k = 1$ A giant component arises, the diameter peaks (sharply), but path lengths are high
- $k > 1$ Almost all nodes will be connected, the component diameter shrinks again, and path lengths shorten

Erdős-Rényi Graph Connectivity (2/3)

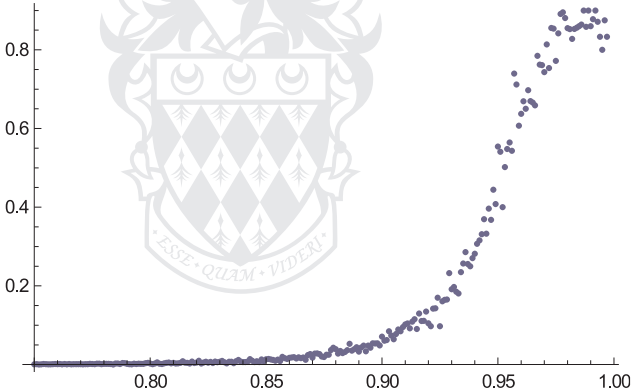


Note that the fraction of edges required to have high probability of the entire graph being connected is much larger ($\Theta(n \log n)$)

Erdős-Rényi Graph Connectivity (3/3)

When studying the **robustness** of a network or graph, one natural question is *what fraction of vertices or edges need to be removed to partition a graph*

Consider vertex removal for $G \in ER(2000, 0.015)$ showing the fraction outside the giant component (vertical axis) plotted against the fraction of vertices removed (horizontal axis):



Network Models

Networks arise in a number of areas, including

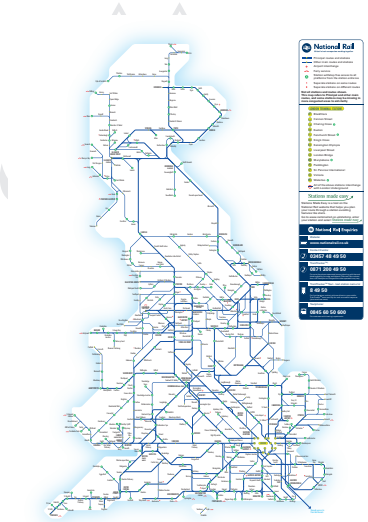
- Social networks such as interactions, relations, but also online networks such as LinkedIn, X, Mastodon, BlueSky, Facebook etc.
- Physical networks, e.g. for transportation
- Large-scale computer networks

Most of these networks are not designed from the ground up, but are they random?

Network Examples

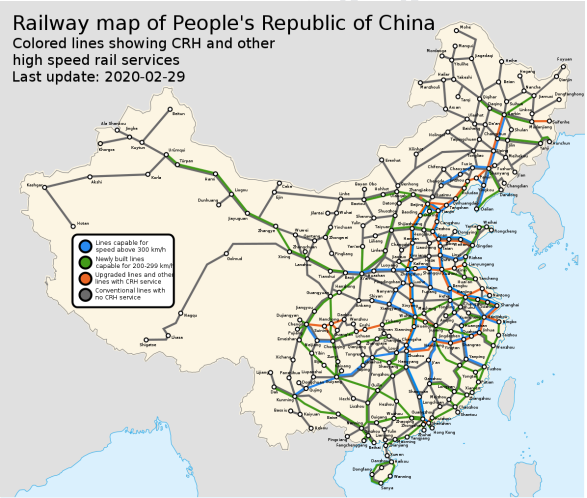
Rail Networks — UK Network (2020)

These networks do not appear very **not** random, and the *degree distribution* does not seem to be Poisson



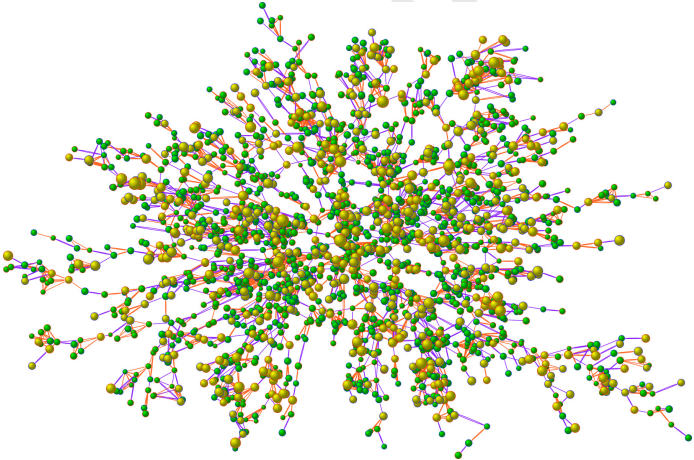
Network Examples

Rail Networks: Chinese High-Speed Network (2020)



Network Examples
Social Networks — Facebook

Social networks are not limited by physical constraints, but are also apparently not random:

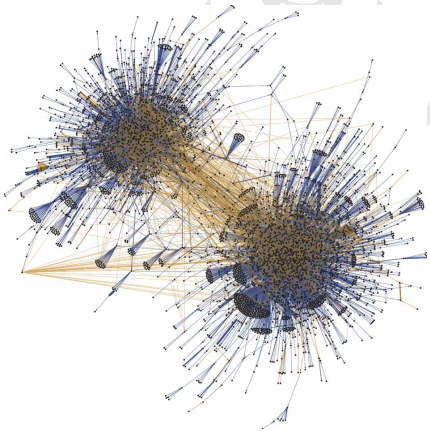


Facebook Network Graph (Source: Facebook)

Network Examples

Social Networks — Twitter/X

X/Twitter network excerpt (here also showing following/retweeting interaction types in colour):



Source: A. Vespignani (Nature Physics 8, 32–39 (2012))

Alternative Network Models

When studying many existing networks, two discrepancies between the properties of Erdős-Rényi graphs and what is observed are of primary concern:

- ER (random) graphs have a low clustering coefficient
- The degree distribution of such networks often does not match with that of ER graphs

These two factors influence other parameters that are important in characterising actual networks, including average path lengths

- Clearly, these in turn influence both graph properties and also e.g. the effectiveness of many types of graph algorithms

The Watts-Strogatz Model

The Small World phaenomenon described by Milgram (1967) highlighted the small values for average path lengths, leading to the popular notion of *Six Degrees of Separation* between any two individuals

Watts and Strogatz (1998) developed an explicit graph construction that seeks to achieve the desired properties by introducing short cuts

- This is achieved by “re-wiring” a regular graph
- A limitation of this construct is that the re-wiring approach implies a fixed number of vertices and also has a relatively homogeneous degree distribution as a side effect

The Watts-Strogatz Algorithm

Watts-Strogatz Algorithm

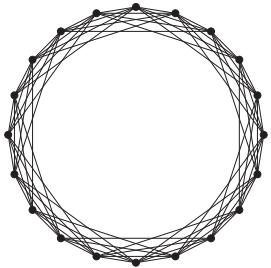
Let $V = \{v_1, v_2, \ldots, v_n\}$ with k even and choose $n \gg k \gg \ln(n) \gg 1$

1. Order the n vertices into a ring (regular graph)
2. Connect each vertex to its first $k/2$ lefthand neighbours, and to its $k/2$ righthand neighbours
3. With probability p, replace edge $\langle u, v\rangle$ with an edge $\langle u, w\rangle$ where $w \neq u$ is randomly chosen, but such that $\langle u, w\rangle \notin E(G)$

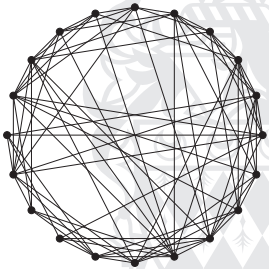
The resulting graphs are denoted as $WS(n, k, p)$-graphs

Examples of Watts-Strogatz Graphs

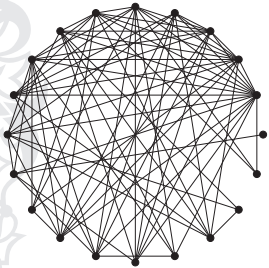
Intuitively, the parameter p provides cross-connections serving as short cuts:



$p = 0.0$



$p = 0.20$



$p = 0.90$

Note on graph size

$n = 20$; $k = 8$; $\ln(n) \approx 3$, so conditions are not fully met

Watts-Strogatz Graph Properties (1/2)

Observation

For many vertices in a WS-graph, $d(u, v)$ will be small:

- Each vertex has k nearby neighbours
- There will be direct links to other "groups" of vertices
- Weak links: The *long* links in a WS-graph that cross the rings seen in the visualisation

Theorem (*Average Path Lengths in W-S Graphs*)

$\forall G \in WS(n, k, 0)$ the average shortest-path length $\bar{d}(u)$ from vertex u to any other vertex is approximated by

$$\bar{d}(u) \approx \frac{(n-1)(n+k-1)}{2kn}$$

Watts-Strogatz Graph Properties (2/2)

Theorem (*Clustering Coefficients in W-S Graphs*)

For any G from $WS(n,k,0)$, $CC(G) = \frac{3}{4}\frac{k-2}{k-1}$.

$WS(n,k,0)$ graphs have relatively long shortest paths, yet high clustering coefficients

However, increasing p shows that average path length drops rapidly when normalising (i.e. dividing by $CC(G_0)$ and $\bar{d}(G_0)$ with $G_0 \in WS(n,k,0)$)

Observations

Observations on the Watts-Strogatz Model (1/2)

The Watts-Strogatz model captures properties that are not satisfied by either random graphs or regular grid-like graphs

- Random graphs have a clustering coefficient of p
- Grid-like graphs exhibit both regularity and locality, but have a high average path length and diameter
- For $p = 0$, a W-S graph will be regular with clustering coefficient approaching $\frac{3}{4}$ for large k and diameter $O(n)$
- For $p = 1$, we have a regular random graph with clustering coefficient p and diameter $O(\log n)$

Observations

Observations on the Watts-Strogatz Model (2/2)

Observation

As noted before, both W-S and random graphs have very similar degrees:

- For random graphs, this would follow a normal distribution around $(n-1)p$
- For W-S graphs, the normal distribution would be around $2k$ (where k is the number of neighbours in the ring)

These degree distributions do not match with many observable networks

Scale-Free Graphs (1/3)

A large body of research investigated physical and logical networks, seeking to understand key properties, including by a group led Barabási (1999)

- One observation in many networks is that the degree distribution is heavily skewed
- Most vertices have a relatively small degree (1–2) with a small number of high-degree nodes where the number of high-degree nodes decreases exponentially rather than following a normal distribution
- Such high-degree vertices are also called *hubs*
- The degree distribution follows a power law distribution, such networks are also called scale-free

Scale-Free Graphs (2/3)

Informally, we can see

Scale-Free Networks (1)

In a scale-free network, $\mathbb{P}[\delta(u) = k] \propto k^{-\alpha}$

More precisely:

Definition (*Scale-Free Network*)

A function f is scale-free if and only if $f(bx) = C(b) \cdot f(x)$, where $C(b)$ is a constant dependent only on b

This raises the question about which network model can generate graphs exhibiting such degree distributions

Scale-Free Graphs (3/3)

Key properties desirable that should also be captured include:

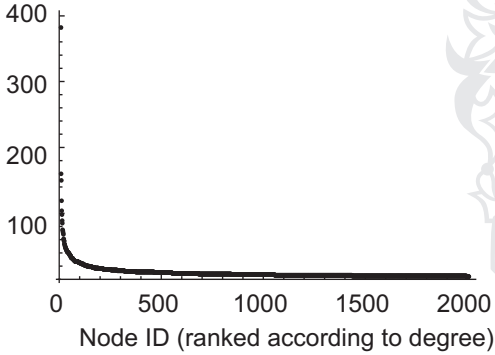
Network Growth Network models allowing to study the growth of a graph over time can provide additional insights over models such as the Watts-Strogatz model

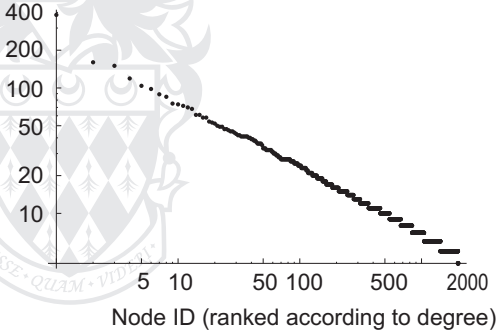
Preferential Attachment Rather than use random edge generation, the model should capture the effect that a high current degree of a vertex increases the probability of a new vertex connecting to it

Scale-Free Graph Distributions

Example of Scale-Free Graph Distribution (1/2)

For a graph with $n = 2000$, the differences in degree among vertices can clearly be seen (linear vs. log scale on vertical):

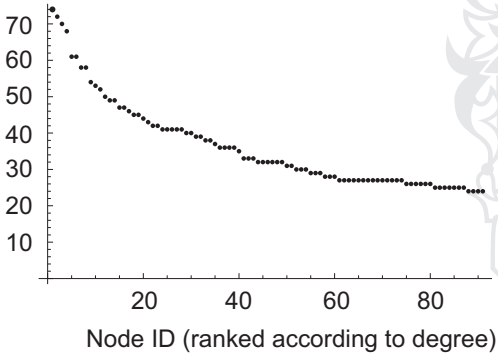


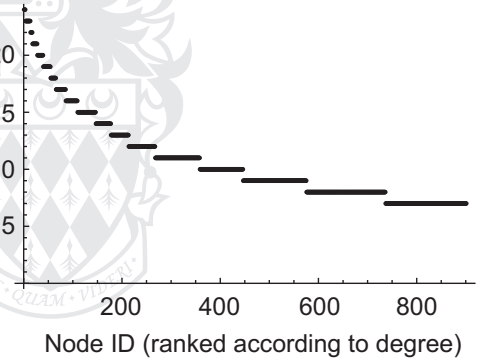


Scale-Free Graph Distributions

Example of Scale-Free Graph Distribution (2/2)

These figures clearly illustrate what is meant by scale-free graphs (graph with 100 vs one with 1000 vertices) :





The Barabási-Albert Model

Algorithm (Barabási-Albert)

$G_0 \in ER(n_0, p)$ with $V_0 = V(G_0)$. At each step $s > 0$:

The Barabási-Albert Model

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$G_0 \in ER(n_0, p)$ with $V_0 = V(G_0)$. At each step $s > 0$:

1. *Add a new vertex $v_s : V_s \leftarrow V_{s-1} \cup \{v_s\}$*

The Barabási-Albert Model

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1. *Add a new vertex $v_s : V_s \leftarrow V_{s-1} \cup \{v_s\}$*
2. *Add $m \leq n_0$ edges incident to v_s and a vertex u from V_{s-1} (and u not chosen before in current step). Choose u with probability*

$$\mathbb{P}[\text{select } u] = \frac{\delta(u)}{\sum_{w \in V_{s-1}} \delta(w)}$$

(Note: Choose u proportional to its current degree)

The Barabási-Albert Model

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3. *Stop when n vertices have been added, otherwise repeat previous two steps*

The Barabási-Albert Model

Algorithm (Barabási-Albert)

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   (Note: Choose u proportional to its current degree)

3. *Stop when n vertices have been added, otherwise repeat previous two steps*

Result: A Barabási-Albert graph, $BA(n, n_0, m)$

Degree Distribution

Degree Distribution in Barabási-Albert Graphs

Theorem

For any $BA(n, n_0, m)$ graph G and $u \in V(G)$:

$$\mathbb{P}[\delta(u) = k] = \frac{2m(m+1)}{k(k+1)(k+2)} \propto \frac{1}{k^3}$$

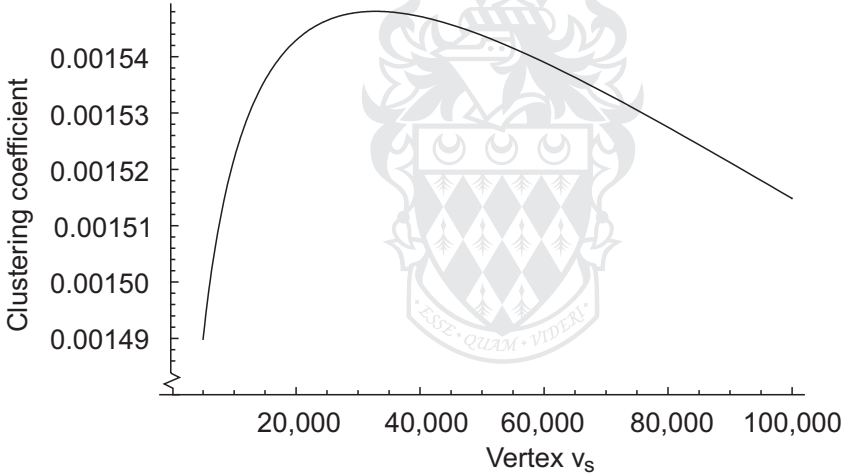
BA-graphs after t steps

Consider clustering coefficient of vertex v_s after t steps in the construction of a $BA(t, n_0, m)$ graph (assume that v_s was added at step $s \leq t$):

$$cc(v_s) = \frac{m-1}{8(\sqrt{t} + \sqrt{s}/m)^2} \left(\ln^2(t) + \frac{4m}{(m-1)^2} \ln^2(s) \right)$$

Clustering in Barabási-Albert Graphs

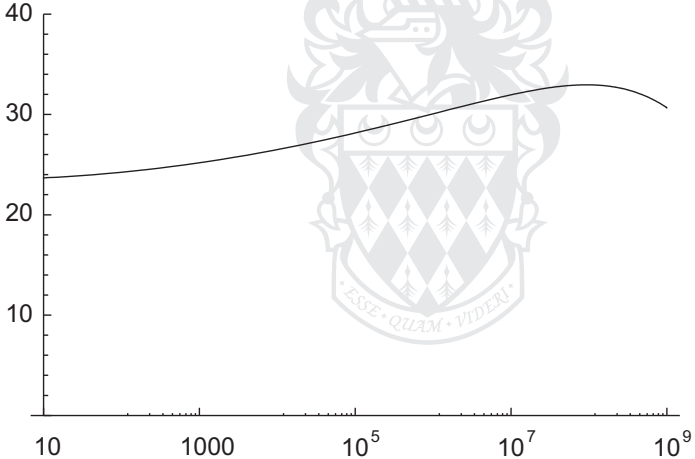
For fixed number of edges being attached m and t steps, consider the clustering coefficient for vertices v_s obtained when varying s:



Comparing Clustering Coefficients

Barabási-Albert and Erdős-Rényi Graph Vertex Cluster Coefficients

Comparing BA and ER graphs, the ratio of $cc(v_s)$ between $BA(N \leq 1\,000\,000\,000, 0,8)$-graph to an $ER(N,p)$-graph shows the much larger cc of BA graphs:



Comparing Average Path Lengths

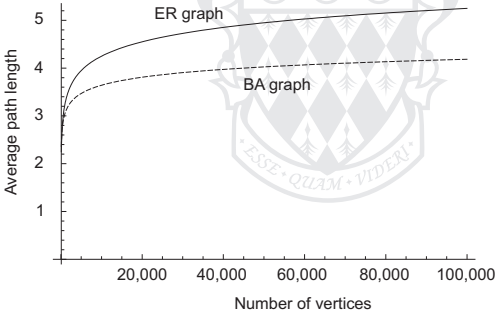
Barabási-Albert and Erdős-Rényi Graph Average Path Lengths

Average Path Lengths

$$\bar{d}(BA) = \frac{\ln(n) - \ln(m/2) - 1 - \gamma}{\ln(\ln(n)) + \ln(m/2)} + 1.5$$

(with $\gamma \approx 0.57772$ the Euler constant)

For the average degree $\bar{\delta}(v) = 10$ we have e.g.:



Summary

This section provided material on

- Properties and Metrics for Characterising Graphs
- Existing Networks and the Erdős-Rényi Model
- The Watts-Strogatz Small-World Model
- Scale-Free Graphs and the Barabási-Albert Model